



GIRRAWEE HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION

2006

MATHEMATICS

*Time allowed - Two hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

Preliminary Learning Outcomes. Mathematics

- P2 provides reasoning to support conclusions which are appropriate to the context.
- P3 performs routine arithmetic and algebraic manipulations involving surds simple rational expressions and trigonometric identities.
- P4 chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques.
- P5 understands the concept of a function and the relationship between a function and its graph.
- P6 relates the derivative of a function to the slope of its graph.
- P7 determines the derivative of a function through the routine application of the rules of differentiation.
- P8 understands and uses the language and notation of calculus.

Question 1(15 marks)

- a) Find, correct to two decimal places:

$$\frac{2.48^2}{\sqrt{3.75 - 2.93}} \quad 2$$

- b) Factorise fully:

(i) $x^3 - 2x^2 - 15x$ (ii) $x^3 + 8$ 4

- c) By expressing $\frac{4}{2 + \sqrt{5}} - \frac{1}{9 - 4\sqrt{5}}$ in its simplest form show that it is a rational number. 3

- d) Express as a single fraction, leaving the denominator in factorised form:

$$\frac{x+5}{x^2+x} + \frac{x+4}{x^2-x} \quad 3$$

- e) Solve the following pair of simultaneous equations:

$$\begin{aligned} 7a + 3b &= 36 \\ 5a + 2b &= 25 \end{aligned} \quad 3$$

Question 2(12 marks)

- a) Solve $|3x - 4| + 2 \leq 7$ 2

- b) Express $0.\dot{6}\dot{4}$ as a fraction in its simplest form. 2

- c) Find the exact value of:

$$49^{\frac{-1}{2}} \times 27^{\frac{2}{3}} \quad 2$$

- d) Solve:

$$x^4 - 10x^2 + 21 = 0 \quad 2$$

- e) Find the value of x if: 2

$$8^{2x-1} = 16^{x+2}$$

- f) Simplify:

$$(\sqrt{2} + 1)^2 + (2\sqrt{3})^2 \quad 2$$

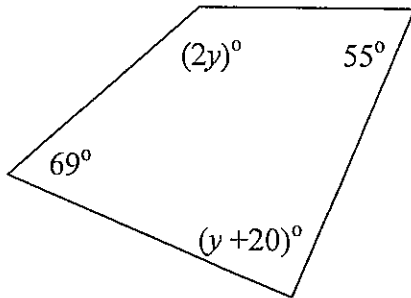
Question 3(12 marks)

a) Find the size of each interior angle of a regular hexagon.

2

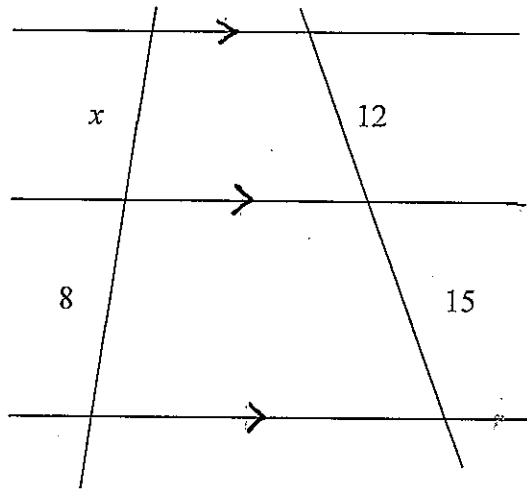
b) Find the value of y , giving reasons

2



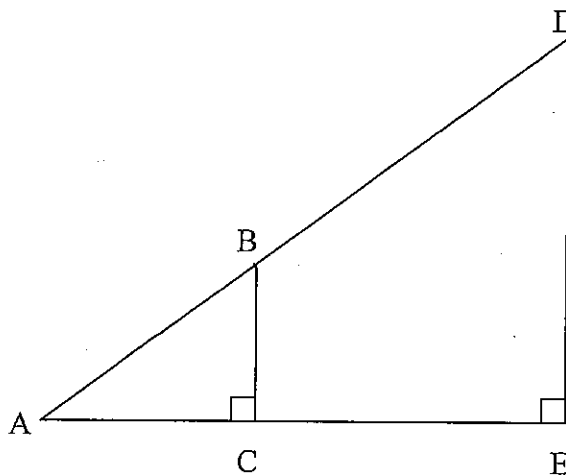
c) Find the value of x , giving reasons. All measurements are in cm.

3



d)

5



$AC = 10\text{cm}$, $CE = 15\text{cm}$, $BC = 8\text{cm}$

(i) Prove that $\triangle ABC \parallel \triangle ADE$.

(ii) Find the length of DE .

Question 4(21 marks)

a) Find the derivative of:

(i) $5x^3 + 4x - 7 + \frac{3}{x^2}$ 2

(ii) $(3x^2 - 7)^{10}$ 2

(iii) $\frac{2x-5}{3x+7}$ 3

(iv) $\sqrt{4-x^2}$ 2

(v) $3x^4(x-4)^5$ 3

b) Find, from first principles, the gradient of the tangent to the curve $y = 16 - x^2$ at the point where $x = -3$ 3c) If $S(t) = 5t^3 - 4t^2 + 5t$, find the value of $S'(-2)$ 2d) Find the equation of the normal to the curve $y = 3 + 6x - 2x^2$ at the point (1, 7). 4**Question 5**(22 marks)

a) A(4, 10), B(-3, 1) and C(5, 7) are the vertices of triangle ABC. Plot the points on a number plane. Find

(i) the coordinates of M, the midpoint of BC 1

(ii) the equation of BC 3

(iii) the perpendicular distance of A from BC 2

(iv) the area of $\triangle ABC$ 2

b) Prove that the points A(3, -2), B(-1, -7) and C(11, 8) are collinear. 2

c) Find the exact value of $\sin 315^\circ$ 2d) Solve: $\sqrt{3} \tan \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ 2e) Prove that $\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$ 3f) ABCD is a parallelogram in which AB = 4cm, BC = 3cm, $\angle ABC = 120^\circ$. Find, in exact form

(i) the area of the parallelogram 2

(ii) the length of the diagonal AC. 3

Question 6(20 marks)

a) Draw separate sketches of the following, showing the important features:

(i) $y = |x - 3|$ 2

(ii) $y = 2^x$ 2

(iii) $y = x^2 - 4$ 2

b) Indicate clearly on a diagram the region determined by the inequality:

(i) $x + 3y - 3 \leq 0$ 3

(ii) $y < x^2 + 2x$ 3

c) Draw a clear sketch of the region which satisfies the inequalities

$x^2 + y^2 \leq 9$ and $y \geq x + 3$ 4

d) (i) Sketch the graph of $y = \frac{1}{2x-1}$ clearly showing any special features. 2

(ii) State the domain and range of the function. 2

Question 7(17 marks)

a) Find the values of a , b and c if

$2x^2 + 3x - 9 \equiv ax(x-1) + b(x-1) + c$ for all x . 3

b) For the parabola

$y = 5x^2 + 10x - 2$, find:

(i) the equation of the axis of symmetry 1

(ii) the coordinates of the vertex 1

c) The quadratic equation $2x^2 + 4x - 3 = 0$ has roots α and β . Find:

(i) $\alpha + \beta$ (ii) $\alpha\beta$ 2

(iii) $\alpha^2 + \beta^2$ (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$ 4

d) For what values of k is the parabola $y = 2x^2 - kx + 8$ positive definite? 2

e) (i) Show that there are two values of k for which the equation

$x^2 + (k-6)x + 2k = 0$ has equal roots. 2

(ii) Determine these roots for each value of k . 2

Year 11 Yearly 2006
Mathematics Solutions

Question 1 (15 marks)

a) 6.79 (2)

b) i) $x^3 - 2x^2 - 15x$
 $= x(x^2 - 2x - 15)$
 $= x(x - 5)(x + 3)$ (2)

ii) $x^3 + 8$
 $= (x + 2)(x^2 - 2x + 4)$ (2)

c) $\frac{4}{2 + \sqrt{5}} - \frac{1}{9 - 4\sqrt{5}}$
 $= \frac{4(9 - 4\sqrt{5}) - (2 + \sqrt{5})}{(2 + \sqrt{5})(9 - 4\sqrt{5})}$

$= \frac{34 - 17\sqrt{5}}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$

$= \frac{34\sqrt{5} + 68 - 85 - 34\sqrt{5}}{-17}$ (3)

d) $\frac{x+5}{x^2+x} + \frac{x+4}{x^2-x}$
 $= \frac{(x+5)(x-1) + (x+4)(x+1)}{x(x+1)(x-1)}$

$= \frac{x^2 + 4x - 5 + x^2 + 5x + 4}{x(x+1)(x-1)}$

$= \frac{2x^2 + 9x - 1}{x(x+1)(x-1)}$ (3)

e) $7a + 3b = 36$ — (1) $\times 2$

$5a + 2b = 25$ — (2) $\times 3$

$14a + 6b = 72$ — (3)

$15a + 6b = 75$ — (4)

(4) - (3) $a = 3$

Sub $a = 3$ into (1)

$7(3) + 3b = 36$

$b = 5$

$\therefore a = 3, b = 5$ (3)

Question 2 (12 marks)

a) $|3x - 4| + 2 \leq 7$

$|3x - 4| \leq 5$

$-5 \leq 3x - 4 \leq 5$

$-1 \leq 3x \leq 9$

$-\frac{1}{3} \leq x \leq 3$ (2)

b) $0.\dot{6}\dot{4}$

$x = 0.646464 \dots$

$100x = 64.646464 \dots$

$99x = 64$

$x = \frac{64}{99}$

(2)

c) $49^{-\frac{1}{2}} \times 27^{\frac{2}{3}}$
 $= \frac{1}{\sqrt{49}} \times (3^3)^{\frac{2}{3}}$

$= \frac{1}{7} \times 9 = \frac{9}{7}$ (2)

d) $x^4 - 10x^2 + 21 = 0$

Let $u = x^2$

$u^2 - 10u + 21 = 0$

$(u - 7)(u - 3) = 0$

$u = 7, 3$

But $u = x^2$

$\therefore x^2 = 7$ or $x^2 = 3$

$x = \pm\sqrt{7}$ or $x = \pm\sqrt{3}$ (2)

e) $8^{2x-1} = 16^{x+2}$
 $(2^3)^{2x-1} = (2^4)^{x+2}$

$\therefore 6x - 3 = 4x + 8$

$x = 5\frac{1}{2}$

(2)

f) $(\sqrt{2} + 1)^2 + (2\sqrt{3})^2$

$= 2 + 2\sqrt{2} + 1 + 12$

$= 15 + 2\sqrt{2}$

(2)

Question 3 (12 marks)

a) $\angle$ sum of hexagon $= 4 \times 180$
 $= 720^\circ$

$\therefore$ vertex angle $= \frac{720}{6}$
 $= 120^\circ$ (2)

b) $2y + y + 20 + 69 + 55 = 360$
 $(\angle \text{sum of quad} = 360^\circ)$

$3y + 144 = 360^\circ$

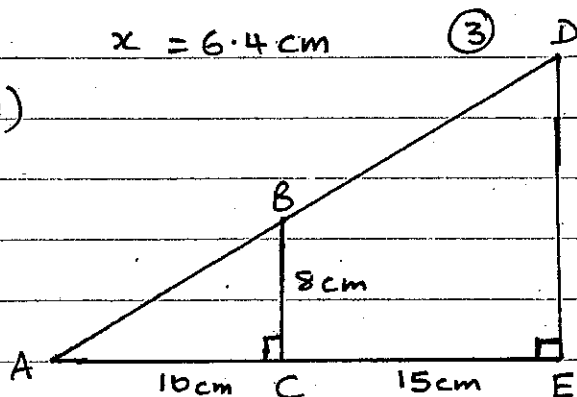
$y = 72^\circ$ (2)

c) $\frac{x}{8} = \frac{12}{15}$ (intercept theorem)

$15x = 96$

$x = 6.4 \text{ cm}$ (3)

d)



i) $\angle BAC = \angle DAE$ (common $\angle$)

$\angle BCA = \angle DEA$ (given)

$\therefore \triangle ABC \sim \triangle ADE$ (equiangular)

(3)

ii) $\frac{DE}{8} = \frac{25}{10}$

$DE = 20 \text{ cm}$ (2)

Question 4 (21 marks)

a) i) $y = 5x^3 + 4x - 7 + 3x^{-2}$

$\frac{dy}{dx} = 15x^2 + 4 - \frac{6}{x^3}$ (2)

ii) $y = (3x^2 - 7)^{10}$

$\frac{dy}{dx} = 60x(3x^2 - 7)^9$ (2)

iii) $y = \frac{2x-5}{3x+7}$

$y' = \frac{vu' - uv'}{v^2}$

$= \frac{2(3x+7) - 3(2x-5)}{(3x+7)^2}$

$= \frac{6x + 14 - 6x + 15}{(3x+7)^2}$

$= \frac{29}{(3x+7)^2}$ (3)

iv) $y = (4-x^2)^{1/2}$
 $y' = \frac{1}{2}(4-x^2)^{-1/2} \cdot -2x$
 $= \frac{-x}{\sqrt{4-x^2}}$ (2)

v) $y = 3x^4(x-4)^5$
 $y' = vu' + uv'$
 $= 12x^3(x-4)^5 + 15x^4(x-4)^4$
 $= 3x^3(x-4)^4(4(x-4) + 5x)$
 $= 3x^3(x-4)^4(9x-16)$ (3)

b) $f(x) = 16 - x^2$
 $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{16 - x^2 - 2xh - h^2 - 16 + x^2}{h}$

$= \lim_{h \rightarrow 0} -2x - h$

$= -2x$

when $x = -3$

$f'(x) = -2(-3) = 6$

$\therefore$ gradient of tangent at $x = -3$ is 6. (3)

$$4) c) S(t) = 5t^3 - 4t^2 + 5t$$

$$S'(t) = 15t^2 - 8t + 5$$

$$S'(-2) = 15(-2)^2 - 8(-2) + 5$$

$$= 81 \quad (2)$$

$$d) y = 3 + 6x - 2x^2$$

$$\frac{dy}{dx} = 6 - 4x$$

$$\text{At } x = 1$$

$$m_{\text{tangent}} = 2$$

$$m_{\text{normal}} = -\frac{1}{2} \text{ pt } (1, 7)$$

Equation of normal:

$$y - y_1 = m(x - x_1)$$

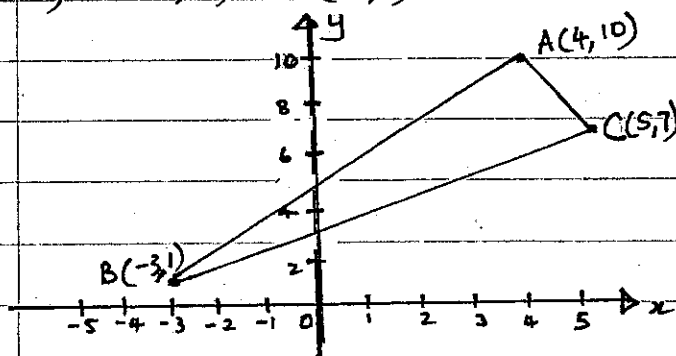
$$y - 7 = -\frac{1}{2}(x - 1)$$

$$2y - 14 = -x + 1 \quad (4)$$

$$x + 2y - 15 = 0$$

Question 5 (22 marks)

$$a) A(4, 10) \quad B(-3, 1) \quad C(5, 7)$$



$$i) M_{BC} = \left(\frac{-3+5}{2}, \frac{1+7}{2} \right)$$

$$= (1, 4) \quad (1)$$

$$ii) m_{BC} = \frac{6}{8} = \frac{3}{4} \text{ pt } (-3, 1)$$

Equation of BC

$$y - 1 = \frac{3}{4}(x + 3)$$

$$4y - 4 = 3x + 9$$

$$3x - 4y + 13 = 0 \quad (3)$$

$$iii) d_{\text{perp}} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3(4) - 4(10) + 13|}{\sqrt{3^2 + 4^2}}$$

$$= 3 \text{ units} \quad (2)$$

$$iv) BC = \sqrt{8^2 + 6^2}$$

$$= 10 \text{ units}$$

$$A_{\Delta ABC} = \frac{1}{2} BC \cdot d_{\text{perp}}$$

$$= \frac{1}{2} \times 10 \times 3$$

$$= 15 \text{ u}^2 \quad (2)$$

$$b) A(3, -2) \quad B(-1, -7) \quad C(11, 8)$$

$$m_{AB} = \frac{-7 + 2}{-1 - 3} = \frac{5}{4}$$

$$m_{BC} = \frac{8 + 7}{11 + 1} = \frac{15}{12} = \frac{5}{4} \quad (2)$$

$\therefore A, B$ and C are collinear

$$c) \sin 315^\circ = \sin (360^\circ - 45^\circ)$$

$$= -\sin 45^\circ$$

$$= -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2} \quad (2)$$

$$d) \sqrt{3} \tan \theta = 1$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ, (180^\circ + 30^\circ)$$

$$\theta = 30^\circ, 210^\circ \quad (2)$$

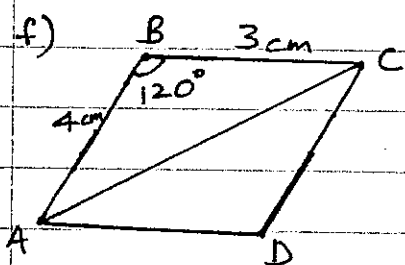
$$e) \tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$$

$$\text{LHS: } \tan \theta + \cot \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta} = \sec \theta \operatorname{cosec} \theta \quad (2) = \text{RHS}$$

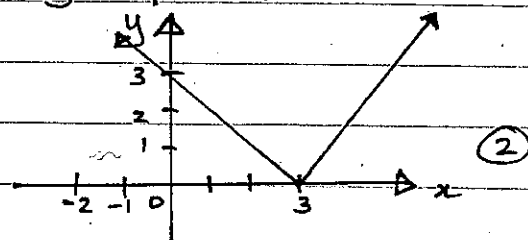


i) $A = 2 \left[\frac{1}{2} \times 4 \times 3 \sin 120^\circ \right]$
 $= 6\sqrt{3} \text{ cm}^2$ (2)

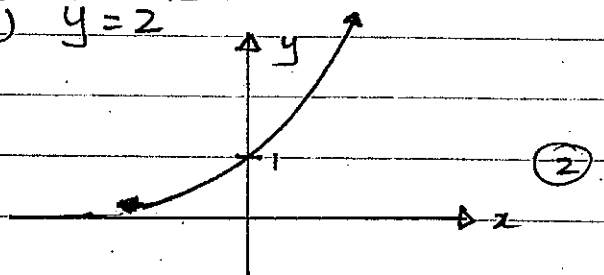
ii) $a^2 = b^2 + c^2 - 2bc \cos A$
 $AC^2 = 3^2 + 4^2 - 2(3)(4) \cos 120^\circ$
 $AC = \sqrt{37} \text{ cm}$ (3)

Question 6 (20 marks)

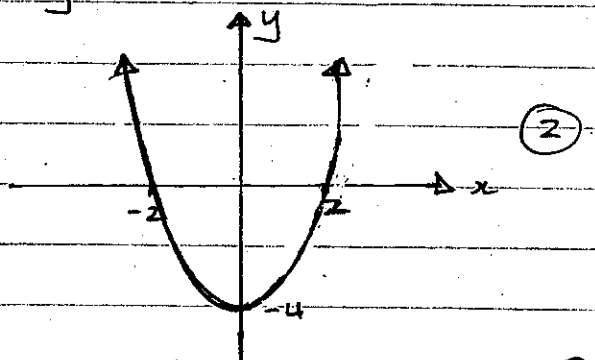
a) i) $y = |x-3|$



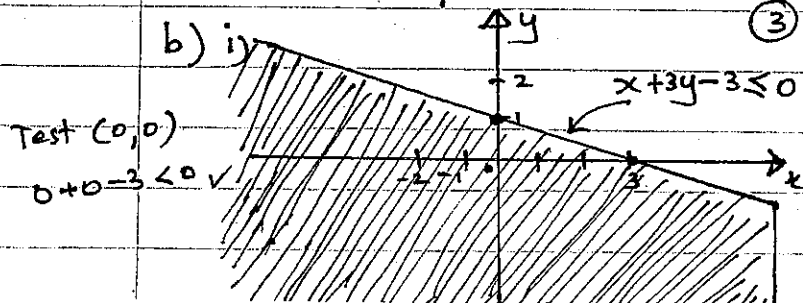
ii) $y = 2^x$



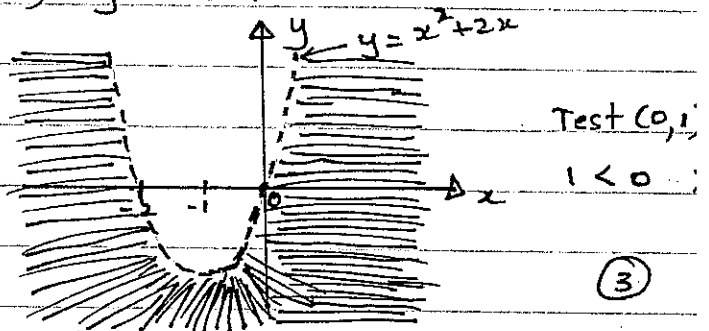
iii) $y = x^2 - 4$



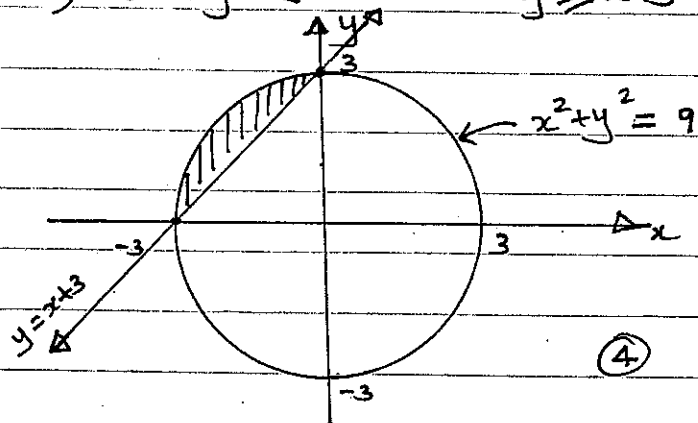
b) i)



ii) $y < x^2 + 2x$

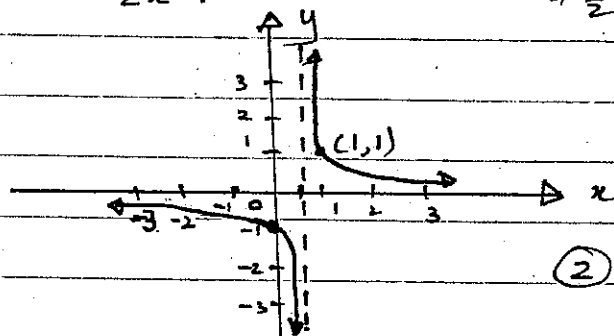


c) $x^2 + y^2 \leq 9$ and $y \geq x+3$



d) i) $y = \frac{1}{2x-1}$

$2x-1 \neq 0$
 $x \neq \frac{1}{2}$



ii) Domain : $x \in \mathbb{R}, x \neq \frac{1}{2}$

Range : $y \in \mathbb{R}, y \neq 0$ (2)

Question 7 (17 marks)

$$\begin{aligned} \text{a) } 2x^2 + 3x - 9 &\equiv ax(x-1) + b(x-1) + c \\ &\equiv ax^2 - ax + bx - b + c \\ &\equiv ax^2 + (b-a)x + c-b \end{aligned}$$

$$\therefore a = 2$$

$$b - a = 3$$

$$b = 5$$

$$c - b = -9$$

$$c = -4$$

(3)

$$a = 2, b = 5, c = -4$$

$$\text{b) } y = 5x^2 + 10x - 2$$

$$\begin{aligned} \text{i) axis of sym: } x &= -\frac{b}{2a} \\ x &= -1 \end{aligned} \quad (1)$$

$$\text{ii) When } x = -1$$

$$y = 5(-1)^2 + 10(-1) - 2$$

$$y = -7$$

$$\therefore \text{Vertex} = (-1, -7) \quad (1)$$

$$\text{c) } 2x^2 + 4x - 3 = 0$$

$$\begin{aligned} \text{i) } \alpha + \beta &= -\frac{b}{a} & \text{ii) } \alpha\beta &= \frac{c}{a} \\ &= -2 & &= -\frac{3}{2} \end{aligned} \quad \begin{matrix} (1) \\ (1) \end{matrix}$$

$$\begin{aligned} \text{ii) } \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= (-2)^2 - 2\left(-\frac{3}{2}\right) \\ &= 7 \end{aligned} \quad (2)$$

$$\text{iv) } \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{-2}{-\frac{3}{2}}$$

$$= \frac{4}{3}$$

(2)

$$\text{d) } y = 2x^2 - kx + 8$$

positive definite : $a > 0, b^2 - 4ac < 0$

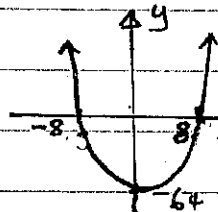
$$(-k)^2 - 4(2)(8) < 0$$

$$k^2 - 64 < 0$$

$$(k-8)(k+8) < 0$$

$$\therefore -8 < k < 8$$

(2)



$$\text{e) i) } x^2 + (k-6)x + 2k = 0$$

equal roots : $b^2 - 4ac = 0$

$$(k-6)^2 - 4(1)(2k) = 0$$

$$k^2 - 12k + 36 - 8k = 0$$

$$k^2 - 20k + 36 = 0$$

$$(k-2)(k-18) = 0$$

$$\therefore k = 2 \text{ or } 18 \quad (2)$$

$\therefore$ there are two values of k

$$\text{ii) When } k = 2$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$\therefore x = 2$$

$$\text{When } k = 18$$

$$x^2 + 12x + 36 = 0$$

$$(x+6)^2 = 0$$

$$\therefore x = -6$$

(2)

$\therefore$ the roots are 2 and -6